Github Link :

Aim : Perform Grey level operation image

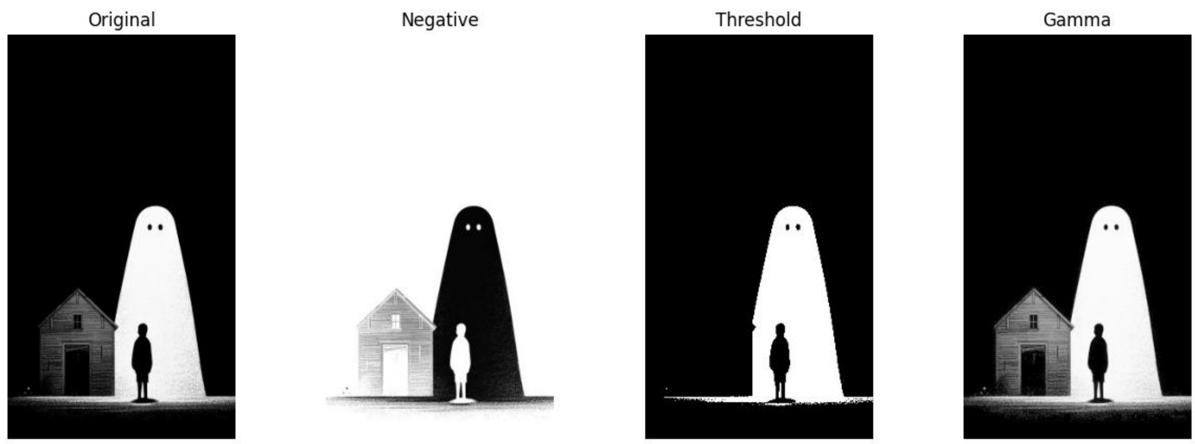
Code :

```
import cv2

import numpy as np

import matplotlib.pyplot as plt


# Read image directly in grayscale

src_image = cv2.imread('/content/28377.jpg', cv2.IMREAD_GRAYSCALE)


# Check if image is loaded

if src_image is None:

    print("Image not found!")

    exit()


# -------------------------

# 1. Image Negation

# -------------------------

negative_image = 255 - src_image


# -------------------------

# 2. Thresholding

# -------------------------

_, thresholded_image = cv2.threshold(src_image, 128, 255, cv2.THRESH_BINARY)


# -------------------------
```

```
# 3. Gamma Correction
# -------------------------
gamma = 1.8

# Normalize image to range [0,1]
normalized_image = src_image / 255.0

# Apply gamma correction

gamma_corrected_image = np.power(normalized_image, 1/gamma)

# Convert back to range [0,255]
gamma_corrected_image = np.uint8(gamma_corrected_image * 255)

# -------------------------
# Display Images
# -------------------------
plt.figure(figsize=(15,5))

plt.subplot(1,4,1)
plt.imshow(src_image, cmap='gray')
plt.title("Original")
plt.axis('off')

plt.subplot(1,4,2)
plt.imshow(negative_image, cmap='gray')
plt.title("Negative")
plt.axis('off')
```

```
plt.subplot(1,4,3)
plt.imshow(thresholded_image, cmap='gray')
plt.title("Threshold")
plt.axis('off')

plt.subplot(1,4,4)
plt.imshow(gamma_corrected_image, cmap='gray')
plt.title("Gamma")
plt.axis('off')

plt.show()
```

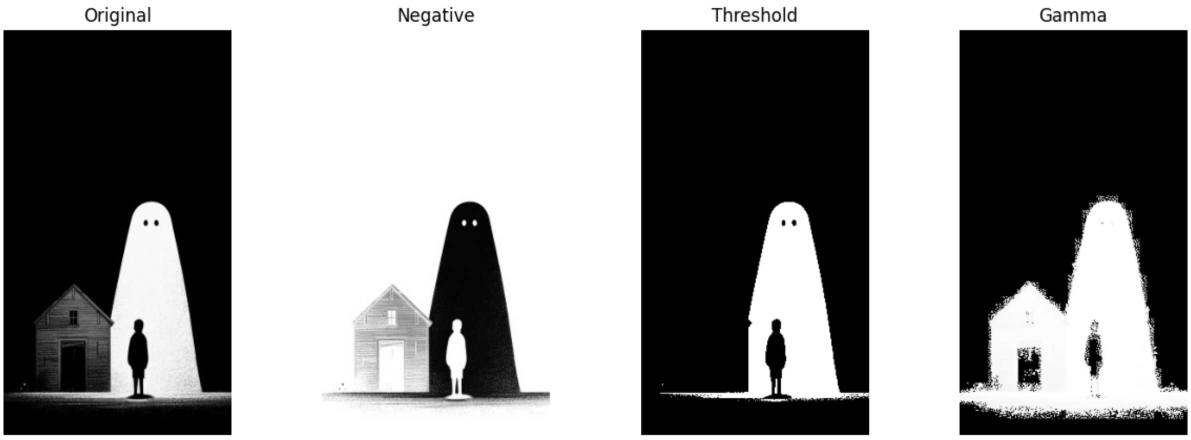
<u>Output :</u>



Input :

gamma Value = 88

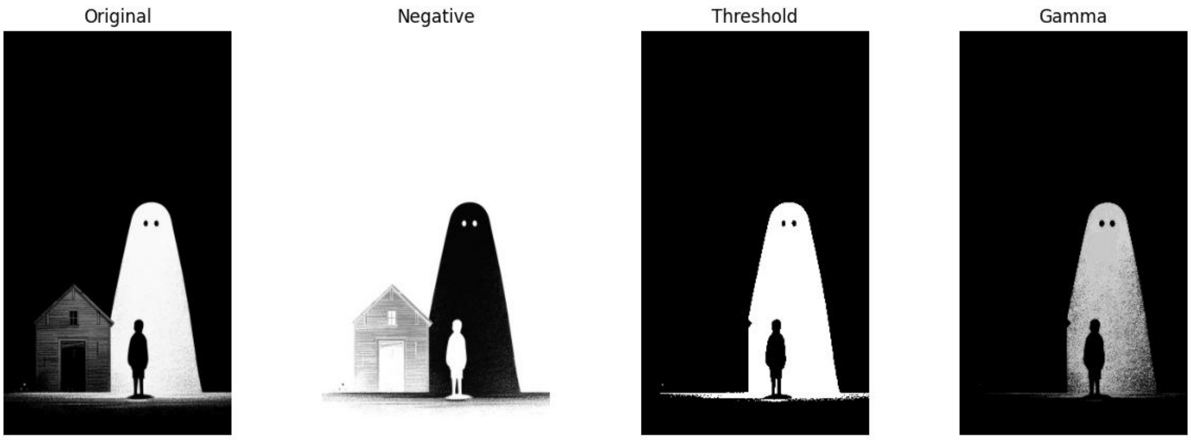
Output :



Input :

gamma value = 0.1

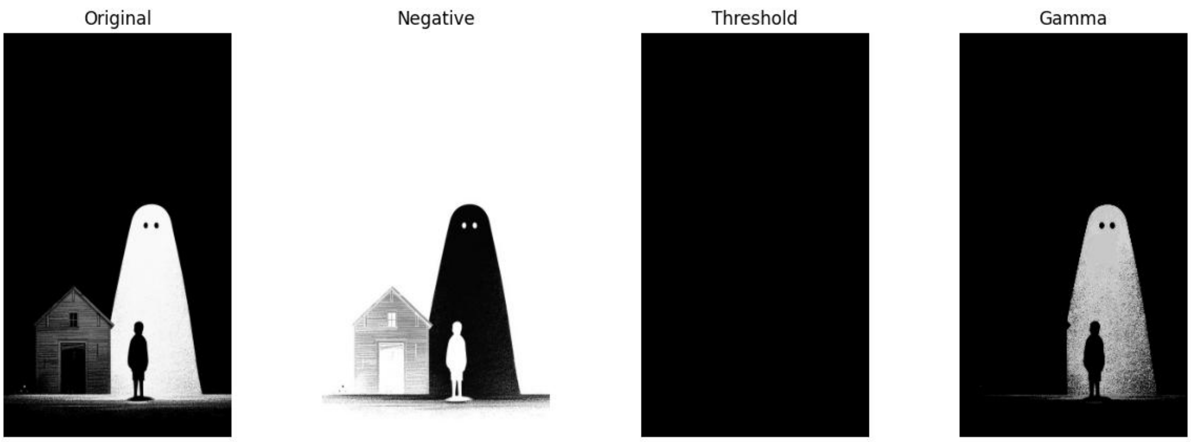
Output :



Input :

Threshold value = 256

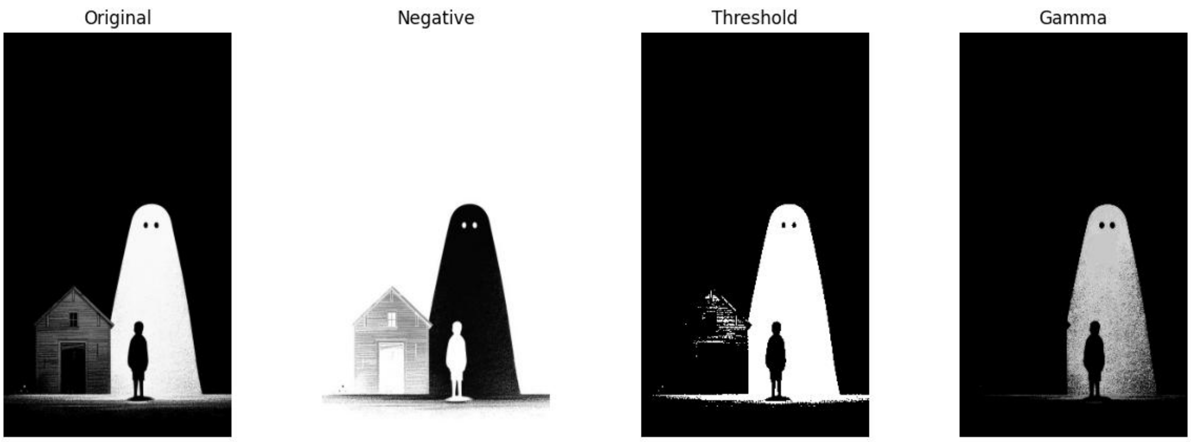
Output :



Input :

Threshold value = 100

Output :



Conclusion :

The program successfully performs basic image processing operations on a grayscale image. It creates a negative image, converts the image into black and white using thresholding, and adjusts the brightness using gamma correction. These techniques help improve image appearance and make important features easier to observe. This experiment demonstrates how simple image processing methods can be applied using Python, OpenCV, and NumPy.